

Belona Sonna

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🐦 [Belonasonna](#)

in [belona-sonna](#)

👤 [belsonna](#)

RESEARCH FOCUS

I am broadly interested in the theory and practice of ethical AI. My current research focuses on **formal verification** of AI ethics principles. I develop **formal explanation based frameworks** to assess fairness, identify proxy discrimination, and audit and repair privacy leakage in AI driven decision making. I am open to collaborative projects in public domains where AI outputs may affect individuals, with a particular emphasis on **healthcare applications**. More information can be found on my [webpage](#).

EDUCATION

- Jan. 2022 - **PhD Candidate, School of Computing, Australian National University**, Canberra, Australia, Ongoing.
Exp Dec. 2026 Thesis: A formal framework based on abductive explanations for assessing and verifying alignment with AI ethics principles in AI driven decision making, with applications to proxy discrimination, individual fairness, and privacy leakage.
- Oct. 2018 - **African Master's in Machine Intelligence (AMMI)**, *African Institute for Mathematical Sciences (AIMS)*, Kigali, Rwanda, Grade: Good.
Feb. 2020 Thesis: Patient flow forecasting in hospitals using ensemble LSTM models.
- Dec. 2013 - **Master of Computer Science, Electronics and Automation, National School of Agro Industrial Science**, Ngaoundere, Cameroon, Grade: Good.
Oct. 2016 Thesis: Identification of color image attributes for monitoring nutrient growth in algal biomass.

SELECTED PUBLICATIONS

- Jan. 2026 **Sonna, B.** Bridging AI and Clinical Reasoning: Abductive Explanations for Alignment on Critical Symptoms. In Proceedings of 2026 Australasian Computer Science Week (ACSW 2026). [Link](#).
- Jan. 2026 **Sonna, B.**, Kouakep, K. T., Ebongue, J. L. F. K. Patient Flow and AI in the Global South: Designing Contextual, Ethical, and Efficient Health Solutions. In C. El Morr, A. Saab, E. Salem-Sokhn (Eds.), AI for Epidemic Preparedness: Innovations and Insights from the Global South. Springer International Publishing (**forthcoming**).

- Dec. 2025 **Sonna, B.**, Grastien, A., Benn, C. Beyond Verification: Abductive Explanations for Post-AI Assessment of Privacy Leakage. Proceedings of the Workshop on Post-AI Formal Methods at AAAI-26. [Link](#).
- Dec. 2025 **Sonna, B.**, Grastien, A. On Explaining Proxy Discrimination and Unfairness in Individual Decisions Made by AI Systems. Proceedings of the Australasian Joint Conference on Artificial Intelligence (AJCAI 2025). [Link](#).
- July 2025 **Sonna, B.**, Momo, A., Grastien, A. Formal Abductive Explanations for Navigating Mental Health Help-Seeking and Diversity in Tech Workplaces. Proceedings of the Empowering Women of Colour in AI-Driven Mental Health Research Workshop at IJCAI-25 (**forthcoming**).
- Jan. 2025 Marmolejo-Ramos, F., Rebecca, M., Florence, J., **Sonna, B.**, et al. Factors Influencing Trust in Algorithmic Decision-Making: An Indirect Scenario-Based Experiment. Frontiers in Artificial Intelligence (2025). [Link](#).

Research Skills

- Skill: **Formal Verification & Abductive Explanations**
 Tools: Z3 Solver, QBF Solver
 Description: Computation of abductive explanations from AI models using SMT translation and Z3 solver; formal verification of AI ethics principles.
- Skill: **Machine Learning & AI Frameworks**
 Tools: Scikit-learn, PyTorch, and Keras
 Description: Training models, translation of models to SMT files for formal reasoning.
- Skill: **Data Analysis & Programming**
 Tools: Pandas, NumPy, Jupyter, Matplotlib, Python
 Description: Handling tabular datasets and empirical validation on real-world datasets.
- Skill: **AI Ethics Assessment**
 Tools: Formal Abductive explanations
 Description: Assessment of proxy discrimination, individual fairness, and privacy leakage in AI-driven decision making.
- Skill: **Research Methodology**
 Tools: Theorems and proofs, Experimental studies
 Description: Integration of theoretical work and empirical studies.

TECHNICAL SKILLS

- O.S: WINDOWS, LINUX, MAC OS
 Database: MySQL, PostgreSQL, Oracle, XAMP, WAMP, Firebase Storage

Tools and Environment Android Studio, Anaconda, Eclipse, Netbeans, Matlab, Webdev, windev, Windev mobile, Spring, OpenCv, Tensorflow, Pytorch, Keras
programming language Python, java, C/C++, Visual Basic, Wlanguage, html/css, PHP

EMPLOYMENT HISTORY

- Feb. 2022 - Tutor of statistical machine learning and Introduction to machine learning
May 2023 courses at the Australian National University, Canberra, Australia, 18 Months.
- Dec. 2020 - Manager, Ethics and Gender Issue at AfroLeadership, Yaounde,
Now Cameroon, Ongoing
- Sept. 2020 - AI Instructor at Vecademy, Yaounde, Cameroon, 9 Months.
May 2021
- Oct. 2019 Research AI Intern at Future Corporation, Tokyo, Working on eye recog-
-Dec.2019 nition and gaze tracking for early detection of Dementia, 2 Months.
- Jul. 2019 - Research Intern at Montreal AI Ethics Institute, My project is on Social
Sept.2019 impact of AI, 2 Months.
- Oct. 2017 - Teaching Assistant of Computer sciences at ISSTMADD(Higher
Jul.2018 Institute of Management Science and Sustainable Development), Ngaoundere, Cameroon, 10 Months.
- Jan. 2017 - Teaching Assistant at the Computer Science Department ENSAI (National
Jul.2017 School of Agro-Industrial Sciences) Ngaoundere, Cameroon, 7 months.

HONOURS AND AWARDS

- Nov.2025 Africa Top 40 under 40 AI Achievers 2025 by Society for AI as Promoter of BEL'S AI Initiative.
- Sep.2025 Best Paper award at the Empowering Women of Colour in AI-Driven Mental Health Research workshop at IJCAI 2025.
- Dec.2021 100 Brilliant women in AI Ethics-2022 list by Women in AI Ethics, as Promoter of the Be Educated and Learn Skills in AI Initiative (BEL'S AI Initiative).
- Oct.2021 80 African women advancing AI in Africa and around the world list by African Shapers, as Promoter of BEL'S AI Initiative.
- Feb.2019 Best project in Building a machine learning-based system for breast cancer prognosis from gene expression data, discerned by Dr. Jean-Philippe Vert, MINES ParisTech and Google AI, during African Master of Machine Intelligence Program.

- Dec.2018 Certificate of Excellence for Best Debate Questions, discerned by Marc Deisenroth, Imperial College UK, during African Master of Machine Intelligence Program.
- Feb.2018 Youth Excellence Award for Technological Innovation by Ministry of Youth and Civic Education of Cameroon.
- Dec.2017 Laureate of the 2nd Edition of the Nanas Awards (first prize): National Competition for the Promotion of Entrepreneurship and Leadership of the Girl, Edition 2017,My project was to build a Mobile App to help people suffering to malnutrition to easily follow their diets.

WORKSHOP AND SEMINARS

- Feb. 2026 Oral presentation at the Adaptive Learning and Intelligent Systems, as part of the Australasian Computer Science Week (ACSW) 2026, Melbourne, Australia.
- Jan. 2026 Oral presentation at the Workshop on Post-AI Formal Methods at AAAI-26, Singapore, Singapore.
- Dec. 2025 Oral presentation at Australasian Joint Conference In Artificial Intelligence 2025, Canberra, Australia.
- Dec. 2025 Poster Presentation Next Generation Responsible AI Symposium 2025, Adelaide, Australia.
- Aug. 2025 Oral Presentation the Empowering Women of Colour in AI-Driven Mental Health Research workshop at IJCAI 2025, Montreal, Online.
- Aug. 2024 Oral presentation Workshop on AI Explainability,Virtual, International Joint Conference on Artificial Intelligence 2024, Jeju, Online.
- Sep. 2023 AI and Machine Learning Summer School 2023,Virtual, Cambridge Centre for AI in Medecine, University of Cambridge, Online.
- Jul. 2020 Attended the Machine Learning Summer School 2020,Virtual, Max Planck Institute for Intelligent systems, Tubingen, Online.
- Dec. 2019 Poster Presentation at Women in Machine Learning Workshop at NeurIPS Conference in Vancouver, Canada.
- August. 2019 DEEP LEARNING INDABA in Kenya
- Apr. 2019 Teaching Assistant in Computer vision Tutorial during INDABA X RWANDA.
- Nov. 2018 Seminar presented by me on Image segmentation at African Master's in Machine Intelligence - AIMS.

CIVIC ENGAGEMENT

- Jul. 2021 Founder of the [BEL'S AI Initiative](#)
- Fev. 2021 Co-organizer of the [WiMLDS Yaounde chapter](#)

Dec. 2020 Manager, Ethics and Gender Issue at [AfroLeadership](#)

Jun. 2020 Member of the [Deep Decisions](#)

REFERENCES

Name: *Dr. Alban Grastien*

Relationship: Primary supervisor of my PhD at the Australian National University.

Contact: ban@grastien.net

Name: *Dr. Michael Norrish*

Relationship: Chair of my PhD supervisory panel at the Australian National University.

Contact: Michael.Norrish@anu.edu.au

Name: *Prof. Lexing Xie*

Relationship: Member of my PhD supervisory panel at the Australian National University.

Contact: lexing.xie@anu.edu.au

Name: *Dr. Jean-Louis Kedieng Ebongue Fendji*

Relationship: Former Computer Science lecturer at the University of Ngaoundere.

Contact: lfendji@gmail.com

LANGUAGES

French: Native

English: Professional